SMA Final Project Proposal

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Project Idea

We are interested in understanding how well modern language models can capture human emotion in social media text. In this project, we will focus on Reddit posts, where users often describe personal experiences and feelings in a natural way. We build directly on the setup introduced in *Fluent but Unfeeling: The Emotional Blind Spots of Language Models* (Shu et al., 2025), which frames emotion prediction as a masked language modeling task. In this setting, the emotion word in each post is hidden, and the model must infer it from the surrounding context. While this sounds straightforward, it turns out to be a challenging problem because emotions are not always expressed directly, and similar situations can elicit different emotional responses.

Our group has six members, so each person will take one model from the comparison presented in the paper and evaluate it individually. This gives us a good spread across model families and sizes. In addition to reproducing the baseline results reported by Shu et al. (2025), we will reproduce baselines in each model's native formulation (MLM for encoder-based models and generative inference for causal or instruction-tuned models) and then fine-tune each model using LoRA to compare against these baselines.

Motivation

The main motivation for this project stems from the paper's findings, which show that even large models still struggle with fine-grained emotion recognition. Although performance improves as model size increases, the gains are not as large as one might expect. This suggests that simply scaling models is not enough to solve the problem fully.

Another reason this problem is interesting is its connection to real-world applications. Emotion understanding plays a role in areas like content moderation, mental health support systems, and general sentiment analysis. If a model misinterprets a user's emotional state, it could lead to poor or even harmful responses. For that reason, improving alignment between model predictions and actual human emotions is important.

Our approach is motivated by the idea that models may benefit more from targeted training than from additional scale. Instead of just using larger models, we want to see whether adapting them specifically for this task yields better results.

Dataset

We will use the EXPRESS data set introduced in the paper. This dataset is constructed from Reddit posts where users explicitly describe their emotions. The dataset is large, with over 30,000 posts, and includes a wide range of emotion labels, well beyond simple categories like positive or negative sentiment.

One of the main advantages of this dataset is that the emotion labels are self-disclosed by users rather than assigned by annotators. This makes the labels more realistic, though it also introduces some variability, since people may describe similar feelings with different words.

Each example in the dataset masks the emotion word in the text, and the model is asked to predict it. This setup allows us to evaluate how well the model understands the context leading to the emotion. For example:

Text: I feel <mask> after failing my exam even though I studied for weeks.
Label: disappointed

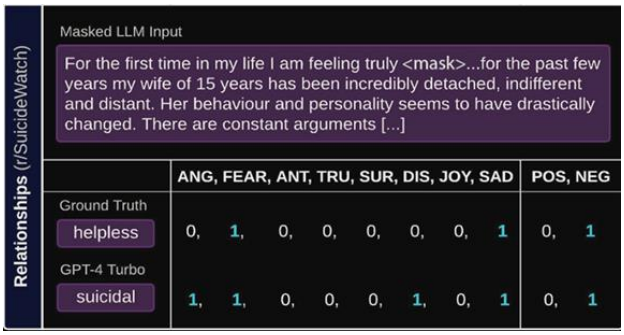

Figure 1: Example of masked emotion prediction and model comparison (Shu et al., 2025).

Approach

We plan to build directly on the authors' codebase, including scripts for dataset handling, model inference, and evaluation. Using their framework ensures that our results are directly comparable to those reported in the paper.

Our work follows two main tracks. First, we reproduce the baseline using the authors' published evaluation artifacts and masked-emotion inference setup. This is done in the model's native formulation, using a masked language modeling (MLM) approach for encoder-based models and generative inference for causal or instruction-tuned models where applicable.

Second, we apply LoRA fine-tuning to adapt each model to the task. LoRA allows us to modify only a small subset of parameters, making training more efficient while still enabling task-specific adaptation. We will use LLaMA Factory to implement this, as it provides a straightforward interface for applying LoRA across different model families.

After fine-tuning, we run inference again and evaluate results under the same pipeline. This allows us to directly compare baseline performance against LoRA-adapted models and determine whether targeted adaptation improves emotion prediction.

Since our group has six members, each person will work with a different model and explore slight variations in setup, such as LoRA configuration or inference strategy. This results in multiple distinct experimental variations across the group while maintaining a consistent evaluation framework. The models used in this study are summarized in Table 1.

If time permits, we may also explore extensions such as QLoRA or other parameter-efficient fine-tuning techniques, as well as testing additional models beyond those included in the original paper. These extensions would help us understand whether further efficiency or model selection choices impact performance.

Metrics

We will use the same evaluation metrics as the paper so that our results are consistent and comparable. These include lexical accuracy, vector accuracy, and F1 score.

Lexical accuracy measures exact matches between predicted and true emotion labels, which is useful but somewhat strict. Vector accuracy is more flexible, as it compares emotions based on underlying dimensions such as joy, fear, and sadness. This is important because two different words can still represent similar emotions. The F1 score helps capture partial correctness and provides a more balanced view of performance. All comparisons are run through the same result-cleaning pipeline and evaluated using AccL, AccV, and F1 metrics to ensure consistency across models and approaches.

Model Family	Model Name	Model Size	Accuracy
RoBERTa	RoBERTa-base	125M	0.37
Longformer	Longformer-base	150M	0.36
RoBERTa	Mental-RoBERTa-base	125M	0.34
Longformer	Mental-Longformer-base	150M	0.33
Flan-T5	Flan-T5-large	780M	0.1
Flan-T5	Flan-T5-xl	3B	0.11
Flan-T5	Flan-T5-xxl	11B	0.17
Gemma2	Gemma-2-2B-it	2B	0.15
Gemma2	Gemma-2-9B-it	9B	0.29
Gemma2	Gemma-2-27B-it	27B	0.34
LLaMA 3.1	LLaMA 3.1-8B-instruct	8B	0.22
LLaMA 3.1	LLaMA 3.1-70B-instruct	70B	0.34
GPT	GPT-3.5-turbo	~175B	0.29
GPT	GPT-4o	~1T	0.39

Table 1: Model size vs emotion detection accuracy across model families (Shu et al., 2025).

References

Hu, E., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, L., & Chen, W. (2021). LoRA: Low-Rank Adaptation of Large Language Models. Retrieved from https://arxiv.org/abs/2106.09685

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